

Atomic MQM · normalized checklist-based scoring

120 figures · 2,252 atoms extracted from ground-truth annotations

(a) What is an atom?

The smallest verifiable claim about a figure, extracted verbatim from the ground-truth annotation.

- The x-axis label is "Number of completions (k)"
- The x-axis uses a logarithmic scale
- The blue solid line starts at approximately 0.42

Critical
Important
Minor

(b) Atom severity caps error severity

Minor atoms can never produce Major errors. Errors against critical atoms inherit the atom's severity.

Atom severity	Complete miss / wrong	Partial miss / slightly off
Critical	Major error	Minor error
Important	Major error	Minor error
Minor	Minor error	Minor error

Critical: chart type, purpose, axis labels, series names. Important: values, ranges, scale types. Minor: colors, styles, tick intervals.

(c) Four-step judge evaluation

Judge receives image + atom checklist + reference description + model description.

01 Accuracy	02 Completeness	03 Hallucination	04 Clarity
atom mentioned but wrong. • numerical value • axis / legend • attribute mapping • structural claim	atom missing entirely. • chart purpose • axis description • visual features	claim not in any atom. • hallucinated content • unwanted interpretation	readability vs reference. • ambiguous • over-generalized • verbose • poor structure

(d) Normalized MQM score

Normalization by atom count guarantees the full 0-100 range independent of figure complexity.

Error weights

Category	Major	Minor
Accuracy	5.0	2.0
Completeness	3.5	1.5
Clarity	2.0	1.0

normalized formula

$$MQM = \max\left(0, 100 - \frac{\sum \text{penalties}}{N_{\text{atoms}} \times 5.0} \times 100\right)$$

penalties include accuracy, completeness, hallucination, clarity

Atom coverage metrics

Atom accuracy	=	atoms with no accuracy error	/	total atoms
Atom completeness	=	atoms mentioned	/	total atoms

Dataset

120 figures · 2,252 atoms · 30 BG · 30 ZH · 39 EN+multi · 21 DE